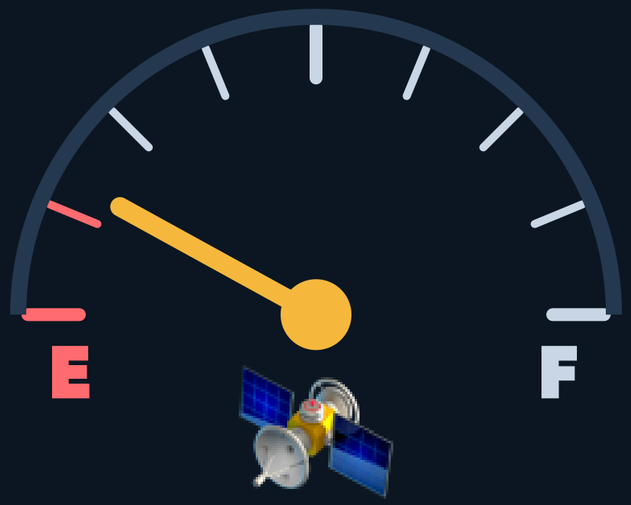
WEEK 6 · DAY 2

When Will It Break? ⛽

Remaining useful life: draw the trend, find the limit, and say **when**, honestly. Then **Quiz 4**. 📝



NIGCOMSAT
AFRICAN ROOTED GLOBALLY POSITIONED

Publica Academy

TODAY'S MISSION · 2 HOURS

Find the limit. Draw the line. Check the call.



Three shapes

how parts grow old



Draw the line

trend to threshold



Check the call

was the battery right?



Quiz 4

forecasting + remaining life

⏱️ 10 min hook

🎤 30 min slides

👩🏾‍💻 45 min hands-on

📝 20 min quiz

🏁 15 min mark + wrap

QUICK QUESTION

“Driver, fuel go reach Ojota?” 🚐



The danfo driver's sum

He doesn't guess. He looks at the gauge, remembers what one trip burns, and divides.

```
litres_left   = 20
litres_a_trip = 4
trips_left    = 20 / 4   # 5 trips
```



That is remaining useful life

How much is left **above the limit**, divided by how fast it is going.

And the smart driver fills up **before** the light comes on, not at Ojota under the bridge. 😅

WHY THIS MATTERS TO NIGCOMSAT

Remaining useful life is a national question 🇳🇬



December 2011

NigComSat-1R launched to 42.5° East



15-year design life

2011 + 15 ≈ 2026



2028 · 2029

targets for NIGCOMSAT-2A and 2B

Will the battery, the amplifiers and the fuel hold until the new ones arrive?
That is a **trend-to-threshold** question.

Today's data: a **synthetic** GEO platform made for teaching. Same questions, made-up numbers. Not NigComSat-1R telemetry.

PART 1

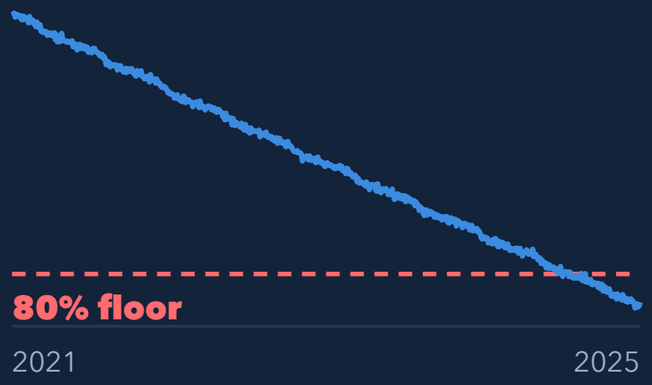
Three ways to grow old 

One satellite, three parts, three limits.

DEGRADATION_ASSETS_DAILY · 1,825 DAYS · 1 JAN 2021 – 30 DEC 2025

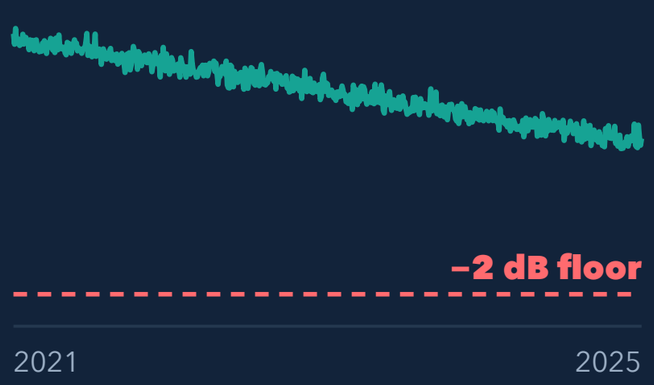
Same satellite, three shapes

🔋 Battery capacity



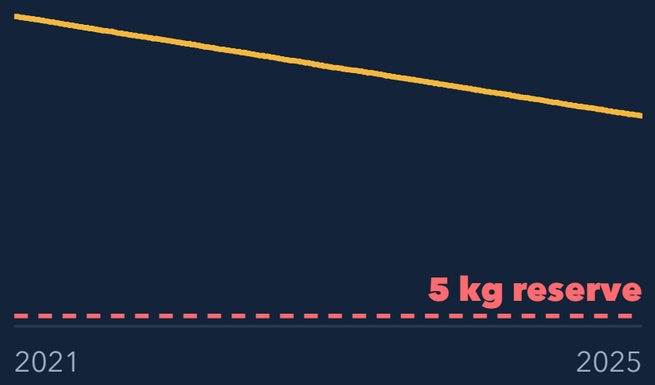
Curves down. **Almost at its floor.**

📡 Amplifier gain



Fades slowly, in a straight line. Far from its floor.

⛽ Propellant



Steps down at every burn. Far from its reserve.

The limits come from `degradation_thresholds.csv`: they are **known up front**, not forecast.

ZOOM IN: PROPELLANT, THE FIRST FOUR MONTHS

A staircase, not a slope 🪜



~14
days between burns

0.39
kg per burn (typical)

~26
burns a year

9.62
kg used a year

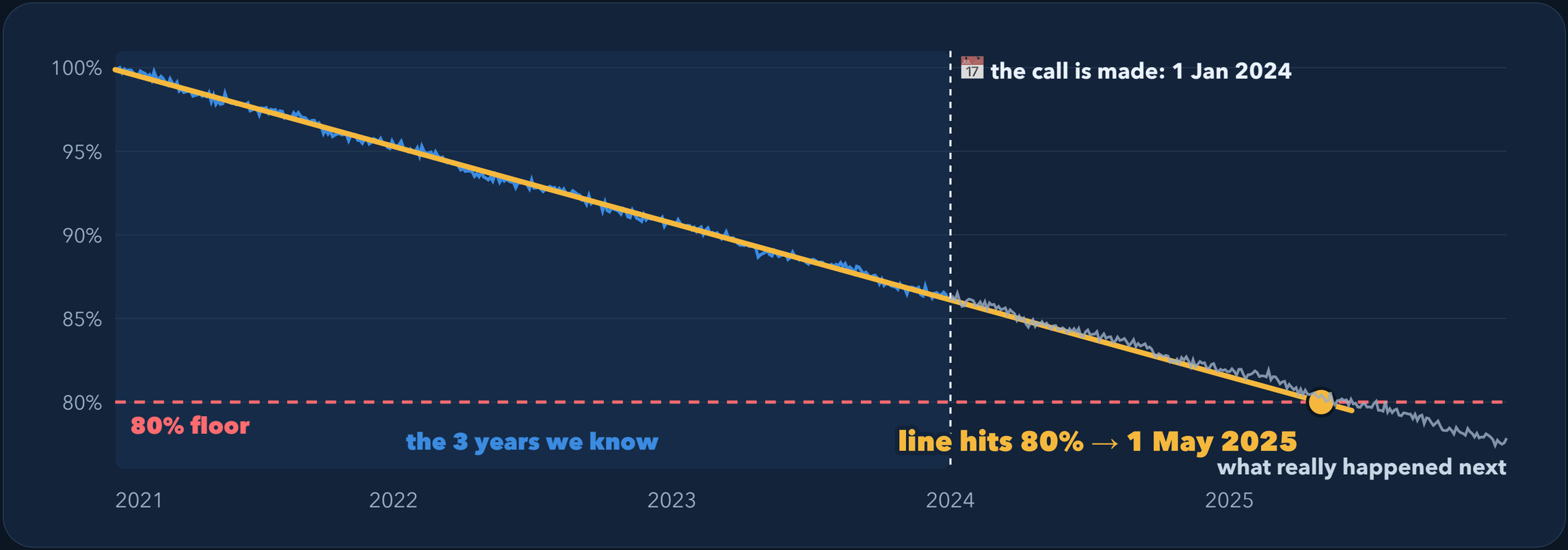
PART 2

Draw the line 📏

Fit the trend on the past. Extend it until it hits the limit.

TREND-TO-THRESHOLD · STAND ON 1 JAN 2024 WITH 3 YEARS OF DATA

Where does the line hit the floor?



Slope: **-4.59% a year**. Remaining useful life on the day of the call: **486 days** (about 16 months).

THE WHOLE METHOD IS TWO SMALL FUNCTIONS

fit_trend + day_of_threshold

```
x = np.arange(len(series))                    # day number 0, 1, 2, ...
slope, intercept = np.polyfit(x, series.values, 1)   # the best straight line

line = np.poly1d([slope, intercept])
future = line(np.arange(len(series) + 4000))          # extend it 4,000 days
crossed = future < threshold                          # True where it is below the limit
day = np.argmax(crossed)                              # the FIRST True
```

`np.polyfit(x, y, 1)` gives **slope and intercept**: two numbers for a straight line.

`np.argmax` of True/False returns the position of the **first True**. Remember that. 👀

THE CODE BOOK'S CALLS · 3 YEARS OF DATA EACH

Two sensible dates… and one strange one



Battery → 80%

1 May 2025



Amplifier → −2 dB

5 May 2033



Propellant → 5 kg

1 Jan 2021

1 Jan 2021 is the **first day of the data**.
The fuel ran out before the satellite even started? 🤔

THE BUG · WEEK 6 CODE BOOK, TOPIC 6.4

np.argmax of “never” is 0 🐞

```
crossed = [False, False, False, False]
np.argmax(crossed)        # 0 !!!

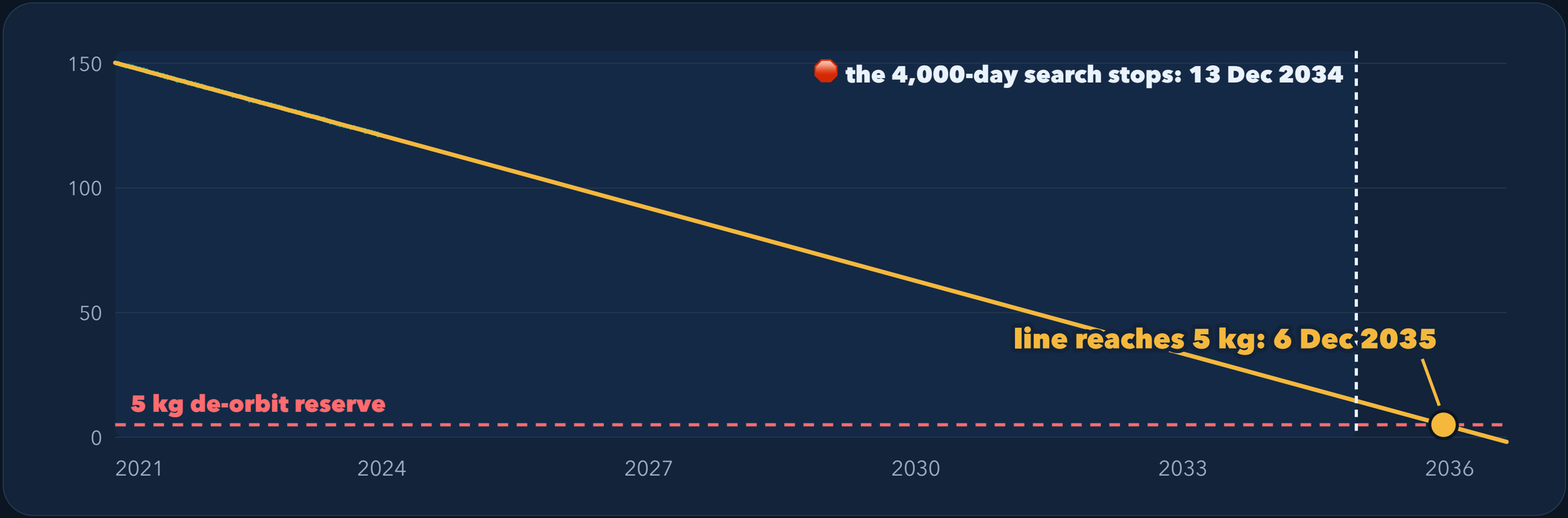
# FIXED in day_of_threshold:
if not crossed.any():
    return None           # "no crossing"
```

The line never reaches 5 kg inside the 4,000-day search (it stops on **13 Dec 2034**). Every value is False, and `argmax` quietly says **0**.

Honest answer: **“no crossing before 13 Dec 2034”**. Then look further.

PROPELLANT · FITTED ON THE SAME 3 YEARS

The line gets there, just later 🔭



Solve the line directly: `day = (5 − intercept) / slope` → **6 Dec 2035**, about 12 years past the data it was fitted on.

THE DANFO DRIVER'S CHECK

The fuel-gauge sum agrees ⛽



96.9 kg ÷ 9.62 = 10.1 years. The line said Dec 2035. **Two methods, one story.**

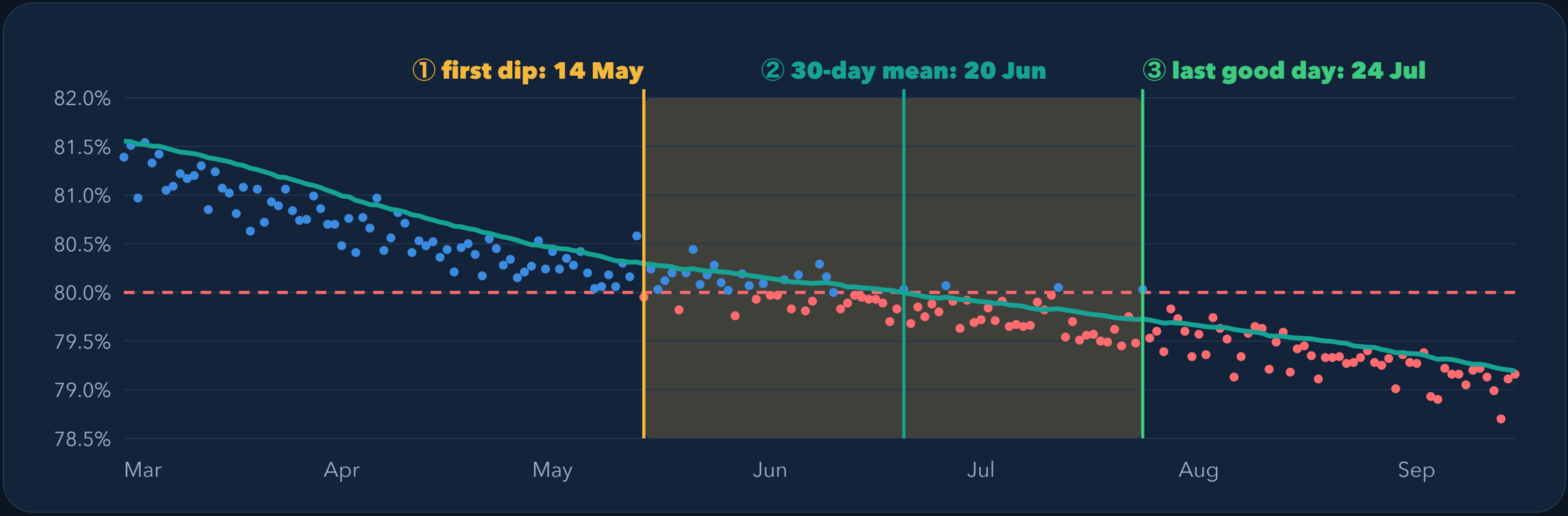
PART 3

Was the battery call right? 🔍

We have the next two years. Let's check.

BATTERY CAPACITY, EVERY DAY OF SPRING AND SUMMER 2025

When did it “cross” 80%? Pick one…



🔵 a day at or above 80% · 🔴 a day below · teal line = 30-day mean · gold = the crossing window

Between the first dip and the last good day, **49 of 72** days are below 80%. A noisy reading doesn't cross on a day: it crosses over a **window**.

THE 3-YEAR CALL WAS 1 MAY 2025

Early by how much? It depends ⚖️

①

vs the first dip (14 May)

13

days early · the code book's answer

②

vs the 30-day mean (20 Jun)

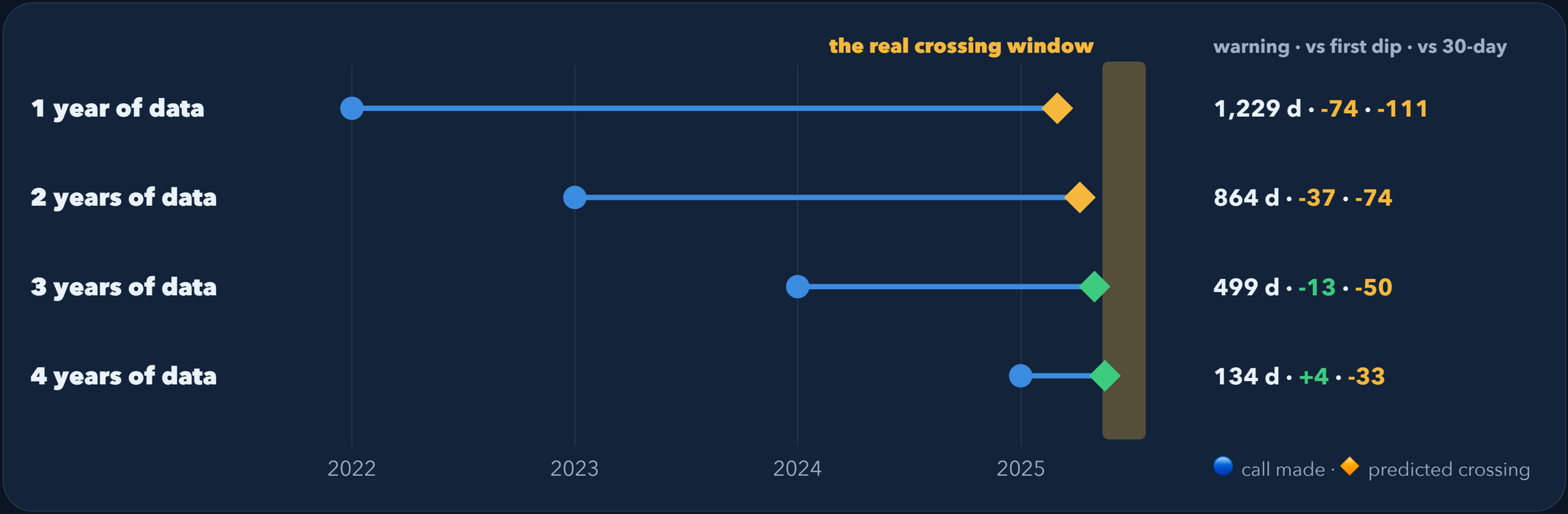
50

days early

Same forecast, two scores. **Define "crossed" before you score**, and write it in your report.

ONE GOOD CALL COULD BE LUCK · CALLBACK TO WEEK 6 DAY 1

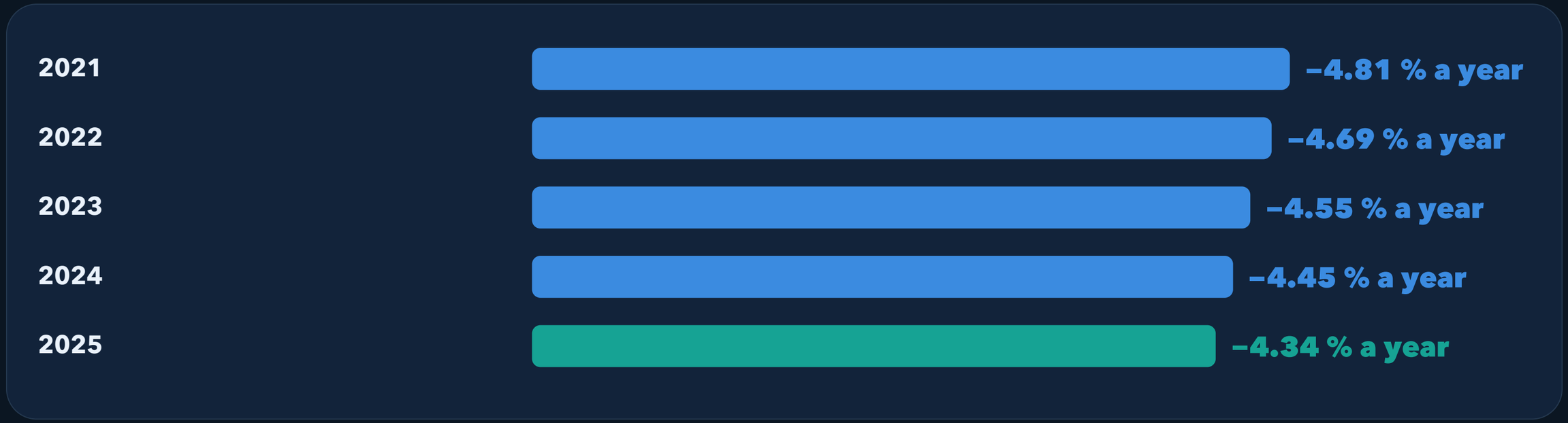
Backtest the METHOD: four starting points



More data → closer call, but **less warning**: 1,229 days of warning with 1 year, only 134 with 4.

WHY DOES THE STRAIGHT LINE CALL IT EARLY?

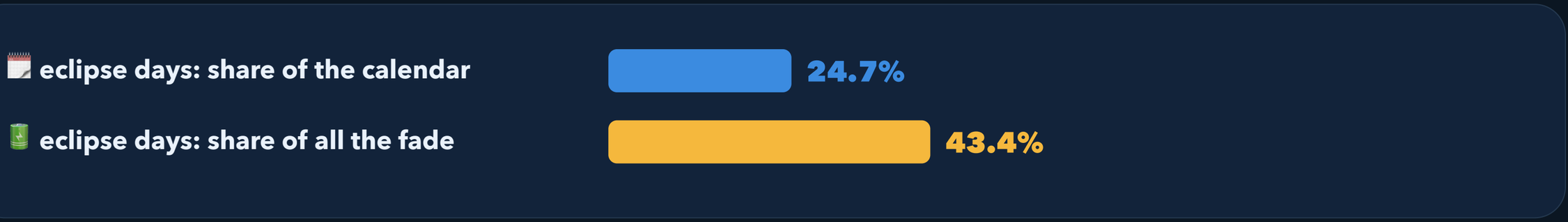
The fade is slowing down 🐢



A line fitted on the early, faster years **overstates** the later fade, so it reaches 80% too soon.
Early is the **safe side** for a battery. But you only know which way you err because you backtested.

LOOK CLOSER AT THE BATTERY

The battery ages when it works 🌒

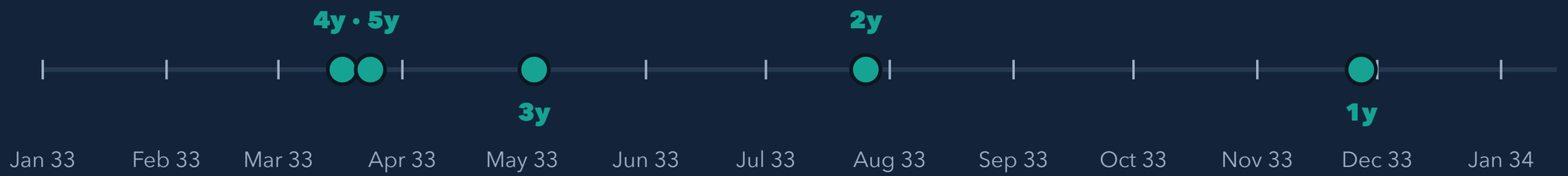


In eclipse season (27 Feb – 13 Apr, 1 Sep – 16 Oct) the battery carries the satellite through the shadow every night,
and loses capacity **2.3× faster** per day. Like your inverter battery in a month of heavy NEPA cuts. 🔌

CALLS MADE FROM 1, 2, 3, 4 AND 5 YEARS OF DATA

The long reach: give a range, not a day 🎯

📡 Amplifier → −2 dB



The calls span about **8 months** of 2033, and even the 5-year call reaches 7.2 years past the data.

⛽ Propellant → 5 kg



Very steady (all within 53 days), because the burns are regular. True only **if the burn plan doesn't change**.

FIVE GOLDEN RULES FOR REMAINING USEFUL LIFE

How a professional says it 👩🏾‍🔧

1. Report a **window**, never a single day
2. Say **how far past the data** you reach
3. Act **before** the crossing
4. **Re-estimate** regularly
5. “No crossing” is an **answer**, not day 0

🔋 **Battery:** first dips below 80% in mid-May 2025, below for good by late July. Act before May.

📡 **Amplifier:** around 2033 (calls span Mar–Nov), 7+ years past our data. Re-check every year.

⛽ **Propellant:** late 2035 to early 2036 if the burn plan holds.

QUIZ 4 · FORECASTING AND REMAINING LIFE

30 questions. 20 minutes. Just you. 

Phones down

notebooks closed, no AI

One letter

circle A, B, C or D (or True/False)

No “always B”

answers are spread across A–D

Mark together

swap papers and tick

Covers baselines (W5D2) · regression + ranges (W5D3) · backtesting (W6D1) · remaining life (today).

YOUR TURN · 45 MINUTES

Call it, check it 🔍

1. Open **W6D2-exercise**, save a copy, run STEP 0
2. **A** Draw the line, find the bug, do the fuel sum
3. **B** Check the battery call from 4 starting points
4. **C** Eclipse ageing, the long reach, the honest report



13 checks to pass

🌶️ Pepper task: bend the line (a curve, degree 2). Where does it land?

Keep the **rule book** open next to you.

TODAY IN 3 LINES

What you now know

📏 **Remaining useful life** = extend the trend until it crosses the known limit. Check “no crossing” honestly.

🔍 **Backtest the method** from several starting points, with “crossed” defined first.

🎯 **Report a window**, say how far you reach, act before the crossing, re-estimate.

Next 👉 **Project clinic & demo day**: five minutes to show your unit what you built. 🎤

Bring your working tool, your report and your honest test table.